14.2

Applying the Basics

18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of $\boxed{200}$ persons who entered the building. ① Do the data in the table indicate that there is a difference in preference for the three entrances? ② Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

CHI $\Rightarrow \chi^2$

① $H_0: p_1 = p_2 = p_3 = \frac{1}{3}$

H_1: At least one entrance preference is different

$E = 200 \times \frac{1}{3} = 66.67$

Entrance	Observed O	expected E
1	83	66.67
2	61	66.67
3	56	66.67

test stat: $\chi^2 = \sum \frac{(O-E)^2}{E}$

$= \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67}$

$= \frac{(16.33)^2 + (-5.67)^2 + (-10.67)^2}{66.67}$

$= 4 + 0.48 + 1.71$

$\chi^2 = 6.19$

$df = k-1 = 3-1 = 2$

$\alpha = 0.05$

$\chi^2_{0.05,2} = 5.991$

$6.19 > 5.991$

reject H_0, we have enough evidence at $\alpha = 0.05$ to conclude that there is a difference in preference among the three entrance

95% CI for entrance 1

$x = 83$

$n = 200$

$\hat{p} = \frac{83}{200} = 0.415$

$CI = \hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$= 0.415 \pm 1.96\sqrt{\frac{0.415(0.585)}{200}}$

$= 0.415 \pm 1.96(0.0348)$

$= 0.415 \pm 0.0683$

$(0.3467, 0.4833)$

$0.3467 < p_1 < 0.4833$, so we are 95% confident that between 34.67% & 48.33% of 200 ppl favor interest 1

22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.[1] In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$H_0: p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = p_7 = \frac{1}{7}$
H_1: the heart attack incidence is not the same for all days

$E = n \times p$
$= 200 \times \frac{1}{7} = 28.5714$

Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Observed	24	36	27	26	32	26	29
Expected	28.5714	28.5714	28.5714	28.5714	28.5714	28.5714	28.5714

test stat

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$\chi^2 = 0.7314 + 1.9314 + 0.0864 + 0.2314 + 0.4114 + 0.2314 + 0.0064$
$\chi^2 = 3.63$
$df = k-1 = 7-1 = 6$

$\chi^2_{0.05,6} = 12.592$

$3.63 < 12.592$ fail to reject H_0

14.3

18. Fatal Accidents Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large
Fatal	67	26	16
Not Fatal	128	63	46

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

	Small	Medium	Large	Row total
Fatal	67	26	16	109
non fatal	128	63	46	237
column total	195	89	62	346

Ho: fatal accident frequency is independent of automobile size
H1: fatal accident frequency is dependent of automobile size

$$E = \frac{(\text{row total})(\text{column total})}{\text{grand total}} = \frac{RC}{G}$$

fatal small: $\frac{(109)(195)}{346} = 61.4306$

fatal medium: $\frac{(109)(89)}{346} = 28.0376$

fatal large $= \frac{(109)(62)}{346} = 19.5318$

not fatal-small $= \frac{(237)(195)}{346} = 133.5694$

not fatal-medium $= \frac{(237)(89)}{346} = 60.9624$

not fatal-large: $\frac{(237)(62)}{346} = 42.4682$

$$\chi^2 = \Sigma \frac{(O-E)^2}{E}$$

$= 0.5050 + 0.1481 + 0.6385 + 0.2322 + 0.0681 + 0.2938 = 1.8857$

$df = (r-1)(c-1) = (2-1)(3-1) = 2$

$\chi^2_{0.05,2} = 5.991$

$1.8857 < 5.991$

fail to reject Ho

there is not enough evidence to conclude that the frequency of fatal accidents depends on the size of automobiles

23. Telecommuting As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

DATA SET DS1403

	Workdays at Home per Week			
Salary	Less Than One	At Least One, but Not All	All at Home	
Under \$50,000	38	16	14	68
\$50,000–\$74,999	54	26	12	92
\$75,000–\$99,999	35	22	9	66
Above \$100,000	33	29	12	74
	160	93	47	300

a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.

b. Use Table 5 in Appendix I to approximate the p-value for this test of hypothesis. Does the p-value confirm your conclusions from part a?

a. Ho: Salary is independent on the number of workdays spent at home
H1: —— dependent —— // ——

$E: \frac{rc}{G}$

Salary	less than one	at least one, but not all	all at home
<50,000	36.2667	21.08	10.6533
50,000-74,999	49.0667	28.52	14.4133
75,000-99,000	35.2	20.46	10.3400
100.000<	39.4667	22.94	11.5933

$X^2 = \Sigma \frac{(O-E)^2}{E}$

$X^2 = 6.4465$

$df = (r-1)(c-1) = 3 \times 2 = 6$

$X^2_{0.05,6} = 12.592$

$6.4465 < 12.592$

fail to reject HO

b. $5.348 < 6.4465 < 7.841$

$0.25 < p < 0.5$

$p = 0.375$

$0.375 > 0.05$

fail to reject HO

14.4

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.[9] Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

DATA SET DS1406

	Married			rate
Educational Level	Yes	No		
Bachelors or higher	187	113	300	62.33%
Some college	162	138	300	54%
High school or less	149	151	300	49.67%
	498	402		

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H0: marrage rates are the same for all educational levels
H1: ——— not the same ———

$E = \frac{RC}{G}$

$\chi^2 = \sum \frac{(O-E)^2}{E}$

$\chi^2 = 10.0611$

$df = (r-1)(c-1) = 2(1) = 2$

$\chi^2_{0.01,2} = 9.210$

$10.0611 > 9.210$ reject H0

marrage rates appear to increase as educational level increases

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

DATA SET DS1407

	Type of Residence			
Family Members	Apartment	Duplex	Single Residence	
1	8	20	1	29
2	16	8	9	33
3	10	10	14	34
4 or More	6	2	16	24
	40	40	40	120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

Ho: The family size distribution is the same for all three residence types
H1: At least one residence type has a different family size distributions
$\alpha = 0.01$

$E_{ij} = \frac{RC}{G}$.

fam member	Apart	Duplex	single res
1	9.67	9.67	9.67
2	11	11	11
3	11.33	11.33	11.33
4/+	8	8	8

test statistic

$X^2 = \Sigma \frac{(O-E)^2}{E}$

$X^2 = 36.499$

$df = (r-1)(c-1) = 2 \times 3 = 6$

$X^2_{0.01,6} = 16.812$

$P \approx 0.0000022$

since $36.499 > 16.812$
&
$p < 0.01$
we reject Ho

there is enough evidence at $\alpha = 0.01$ to conclude that family size distributions differ significantly among aparts, duplexes & single residences